Cloud Computing
Lab 2

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SE2

Task 1: Redis Replication and CAP

1. Connect to redis-node1 and redis-node2 using redis-cli.

```
Last login: Sat Apr 25 23:26:31 on ttys003
mariamibrahim@mariams-air distributed-lab % nano docker-compose.yml
mariamibrahim@mariams-air distributed-lab % docker-compose up -d
docker ps
[+] Running 25/17
 ✔ redis-node2 Pulled                                          52.5s
 ✔ etcd Pulled                                                 21.7s
 ✔ redis-node1 Pulled                                          52.5s

[+] Running 4/4
 ✔ Network distributed-lab_default  C...                        0.0s
 ✔ Container redis-node1            Started                     0.9s
 ✔ Container redis-node2            Started                     1.0s
 ✔ Container etcd                   Started                     1.0s
CONTAINER ID   IMAGE                      COMMAND                  CREATED        STATUS                  PORTS                                          NAMES
fc344c543a99   redis:7                    "docker-entrypoint.s…"   1 second ago   Up Less than a second   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp    redis-node1
3265b11931bb   redis:7                    "docker-entrypoint.s…"   1 second ago   Up Less than a second   0.0.0.0:6380->6379/tcp, [::]:6380->6379/tcp    redis-node2
46fff39819f5   quay.io/coreos/etcd:v3.5.9 "etcd --advertise-cl…"   1 second ago   Up Less than a second   0.0.0.0:2379->2379/tcp, [::]:2379->2379/tcp    etcd
0f2532898019   nginx                      "/docker-entrypoint.…"   2 hours ago    Up 2 hours              0.0.0.0:8080->80/tcp, [::]:8080->80/tcp        focused_golick
mariamibrahim@mariams-air distributed-lab %
```

2. Write a key to redis-node1 and quickly read from redisnode2.

```
mariamibrahim@mariams-air distributed-lab % docker exec -it redis-node1 redis-cli SET mykey "hello from node1"
OK

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redis-node1
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab % docker exec -it redis-node2 redis-cli GET mykey
"hello from node1"

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redis-node2
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab %
```

Node 2 Could read the key “hello from node1”

3. Simulate network partition by stopping redis-node2 temporarily.

```
mariamibrahim@mariams-air distributed-lab % docker stop redis-node2
redis-node2
mariamibrahim@mariams-air distributed-lab % docker exec -it redis-node1 redis-cli SET mykey "written during partition"
OK

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redis-node1
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab % docker start redis-node2
redis-node2
mariamibrahim@mariams-air distributed-lab % sleep 3
mariamibrahim@mariams-air distributed-lab % docker exec -it redis-node2 redis-cli GET mykey
"written during partition"

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redis-node2
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab %
```

- Node 2 stopped
- Redis kept accepting writes
- node2 auto-resynced
- Data was not lost

Task 2: Raft with etcd

```
mariamibrahim@mariams-air distributed-lab % docker exec -it etcd etcdctl put foo bar
OK

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug etcd
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab % docker exec -it etcd etcdctl get foo
foo
bar

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug etcd
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab % docker exec -it etcd etcdctl endpoint status --write-out=table
+----------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+
|    ENDPOINT    |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+----------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+
| 127.0.0.1:2379 | 8e9e05c52164694d |   3.5.9 |   20 kB |      true |      false |         2 |          5 |                  5 |        |
+----------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug etcd
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab %
```

```
mariamibrahim@mariams-air distributed-lab % docker stop etcd
etcd
mariamibrahim@mariams-air distributed-lab % docker exec -it etcd etcdctl get foo

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug etcd
    Learn more at https://docs.docker.com/go/debug-cli/
Error response from daemon: container 46fff39819f52dae88c2791a4b01300a6677dd2997b5793a6a9c04eb4ee77609 is not running
mariamibrahim@mariams-air distributed-lab % docker start etcd
sleep 3
docker exec -it etcd etcdctl endpoint status --write-out=table
etcd
+----------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+
|    ENDPOINT    |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+----------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+
| 127.0.0.1:2379 | 8e9e05c52164694d |   3.5.9 |   20 kB |      true |      false |         3 |          7 |                  7 |        |
+----------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug etcd
    Learn more at https://docs.docker.com/go/debug-cli/
mariamibrahim@mariams-air distributed-lab %
```

- RAFT TERM went from 2 → 3 — that's proof that Raft detected the leader was gone and held a new election automatically